

Ameen Ismail

☎ +1 (607) 379-5917 • ✉ ameenismail@uchicago.edu • 🌐 ameenismail.github.io

Summary

- Theoretical particle physicist with strong analytical skills interested in quantitative research
- Published 18 papers, building models and developing analytical and computational tools to address open problems
- Skills: Python (NumPy, SciPy, Matplotlib), bash, Linux, L^AT_EX, Mathematica

Education

Cornell University

Ph.D. in Physics (research focus: particle physics beyond the Standard Model)

Ithaca, NY

2018–2024

Carleton University

B.Sc. Honours with High Distinction, Biology and Physics, Minor in Mathematics

Ottawa, ON

2014–2018

Experience

University of Chicago

Oehme Postdoctoral Fellow

Chicago, IL

2024–Present

- Pioneered novel paradigms to explain the origin of the Higgs boson mass, resulting in 3 papers, 1 PRL with 59 citations [1]
- Introduced new analytical techniques to study the phenomenon of confinement, resulting in 3 papers, including [2]
- Studied signatures of cosmological phase transitions, e.g. [3], inferring parameters with Markov chain Monte Carlo methods
- Presented my research in 22 invited talks and 11 contributed talks at leading universities, workshops, and conferences
- Mentored 3 graduate students, collaborating on projects to write their first papers together

Cornell University

Graduate student

Ithaca, NY

2018–2024

- Constructed models of dark matter and tested them against experimental data, resulting in 6 papers, 1 single-author [4]
- Developed GMCALCv1.5 to calculate observables in a class of models and compare them to 1000+ experimental searches
- Collaborated in a large group (100+) to propose a new experimental facility, resulting in two white papers
- Taught two undergraduate and three graduate physics classes as a teaching assistant

Awards & Honors

- 2021–2024 NSERC PGS D, Government of Canada fellowship providing 3 years of full funding
- 2021 NSERC CGS D, offered to top-ranked PGS applicants, declined due to being untenable outside Canada
- 2020–2021 Cornell Fellowship, full fellowship awarded to top 20% of Cornell doctoral students
- 2018–2019 Presidential Life Sciences Fellowship, full fellowship awarded to 10 incoming Cornell doctoral students
- 2018 NSERC CGS M, one-year full fellowship to support M.Sc. studies, declined to pursue Ph.D.
- 2017 C.A.B. Betts Memorial, awarded annually to one Carleton physics student
- 2016 Trevor A. Harwood Memorial, awarded annually to one Carleton physics student
- 2016–2018 NSERC Undergraduate Student Research Awards, annual award funding summer research internships

Selected Publications

Full list: <https://inspirehep.net/authors/1797161>. Authors are listed alphabetically in my field.

- [1] Csáki, C., D’Agnolo, R. T., Geller, M., **Ismail, A.**, “Crunching Dilaton, Hidden Naturalness”. In: *Phys. Rev. Lett.* 126 (2021), p. 091801.
- [2] Csáki, C., **Ismail, A.**, “Holographic energy correlators for confining theories”. In: *JHEP* 11 (2024), p. 140.
- [3] Csáki, C., Geller, M., Heller-Algazi, Z., **Ismail, A.**, “Relevant dilaton stabilization”. In: *JHEP* 06 (2023), p. 202.
- [4] **Ismail, A.** “Ultraheavy FIMP dark matter and conformal sectors”. In: *Phys. Rev. D* 108.9 (2023), p. 095049.